

Course Code	304-05																																																						
Course Title	AI-Oriented Programming and Data Structures in Python (A.I.O.P.D.)																																																						
Credits	4																																																						
Course Category	Major Course																																																						
Level of Course	300-399 (Higher Level)																																																						
Teaching per Week	3 Hours Theory + 2 Hours Practical work																																																						
Minimum weeks per Semester	15 (Including class work, examination, preparation etc.)																																																						
Review / Revision	-																																																						
Implementation Year:	A.Y. 2025-2026																																																						
Course Objective	OB1: To introduce the fundamentals of Object-Oriented Programming using Python, with emphasis on concepts like encapsulation, inheritance, and polymorphism, and their relevance in AI-based programming. OB2: To develop a strong foundation in core data structures such as stacks, queues, trees, and sorting algorithms essential for efficient data handling and AI problem-solving. OB3: To equip students with AI problem-solving methodologies including uninformed and informed search strategies, logic representation, and inference mechanisms. OB4: To enable hands-on implementation of AI algorithms and techniques using Python libraries and tools through practical exercises and mini projects. OB5: To foster an understanding of real-world AI applications through the study of expert systems, decision trees, and case studies on widely used AI tools like ChatGPT, Teachable Machine, and IBM Watson.																																																						
Pre-requisite	Basic understanding of Python programming, including variables, control structures, and functions. Familiarity with elementary statistics, logic, and problem-solving techniques will be beneficial for understanding core AI concepts and data structure implementations.																																																						
Course Outcomes	CO1: <i>Describe</i> object-oriented programming principles and linear/non-linear data structures using Python. (Remembering, Understanding) CO2: <i>Implement</i> stacks, queues, sorting algorithms, and trees to solve structured computational problems in AI. (Applying) CO3: <i>Apply</i> various uninformed and informed AI search strategies to model and solve state-space and constraint satisfaction problems. (Applying, Analyzing) CO4: <i>Analyze</i> knowledge representation schemes and logic-based reasoning for intelligent decision-making systems. (Analyzing, Evaluating) CO5: <i>Develop</i> practical AI solutions using Python-based tools and libraries by integrating rule-based systems, logic engines, and modern AI platforms. (Creating)																																																						
Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO)	<table border="1"> <thead> <tr> <th></th><th>PSO1</th><th>PSO2</th><th>PSO3</th><th>PSO4</th><th>PSO5</th><th>PSO6</th><th>PSO7</th><th>PSO8</th></tr> </thead> <tbody> <tr> <td>CO1</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>CO2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>CO3</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>CO4</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>CO5</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr> </tbody> </table>		PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PSO7	PSO8	CO1									CO2									CO3									CO4									CO5								
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Course Content	Unit-1 : Basics of Object Oriented Programming: 1.1 Concepts of Class and Object in Python (Defining classes, objects, and the self-keyword) 1.2 Data Encapsulation and Inheritance:																																																						

- 1.2.1 Access control concepts (Public, Private, Protected) and their difference in Python
- 1.2.2 Declaring simple class, member variables, and member functions in Python
- 1.2.3 Concepts of Data hiding, abstraction, and encapsulation in Python.
- 1.2.4 Concepts of Inheritance in Python and types of inheritance (Single, Multiple, Multilevel)
- 1.2.5 Constructors (__init__) and Destructors (__del__) in Python
- 1.3 Polymorphism:
 - 1.3.1 Concepts of Polymorphism in Python
 - 1.3.2 Compile-time and run-time Polymorphism in Python
 - 1.3.3 Method Overloading and Overriding in Python: Concepts, difference, and application
- 1.4 Introduction to Data Structures and their application areas in Python (using lists, sets, tuples, dictionaries)
 - 1.4.1 Difference between Linear and Non-Linear Data Structures in Python

Unit-2 : Data Structure fundamentals for A.I.:

- 2.1 Stack:
 - 2.1.1 Concepts of Stack (LIFO)
 - 2.1.2 Push, Pop, and Display operations
 - 2.1.3 Application areas of Stack in Python (Infix to Postfix conversion, Postfix to Infix)
- 2.2 Queue:
 - 2.2.1 Concepts of Queue (FIFO)
 - 2.2.2 Implementation of Queue in Python:
 - 2.2.2.1 Simple Queue: insert, delete, and display using lists
 - 2.2.2.2 Double-ended Queue (Deque): insert, delete, and display using collections.deque
- 2.3 Understanding Sorting algorithms:
 - 2.3.1 Concepts of sorting and its time complexity
 - 2.3.2 Implementation of Selection Sort, Bubble sort and Quick sort using Python library functions and modules.
- 2.4 Concepts of Trees:
 - 2.4.1 Definition of a tree, root, leaf, and internal nodes
 - 2.4.2 Concepts of Binary Tree, Binary Search Tree (BST) and AVL Tree

Unit-3: AI Problem Solving Techniques

- 3.1 Problem Formulation in AI
 - 3.1.1 State-space representation
 - 3.1.2 Components: Initial state, goal state, operators
- 3.2 Uninformed Search Strategies
 - 3.2.1 Breadth-First Search (BFS)
 - 3.2.2 Depth-First Search (DFS)
 - 3.2.3 Uniform Cost Search
- 3.3 Informed Search Strategies
 - 3.3.1 Heuristics and Evaluation Functions
 - 3.3.2 Best-First Search, A* Search
- 3.4 Constraint Satisfaction Problems (CSP)
 - 3.4.1 Variables, domains, constraints
 - 3.4.2 Backtracking and forward checking
- 3.5 Understanding Intelligence through Logic and Knowledge
 - 3.5.1 Knowledge Representation Techniques
 - 3.5.2 Propositional logic
 - 3.5.3 First-order logic
 - 3.5.4 Semantic networks and frames
- 3.6 Reasoning and Inference
 - 3.6.1 Inference rules and resolution
 - 3.6.2 Forward and backward chaining

	UNIT-4: Practical Applications of AI using Python 4.1 Rule-Based Expert Systems 4.1.1 Architecture of expert systems 4.1.2 Case study: MYCIN or any rule-based system 4.2 Implementing Search Algorithms using Python Libraries 4.2.1 BFS and DFS using Python 4.2.2 A* Search implementation 4.3 Implementing Logic and Rule-Based Systems 4.3.1 Creating a simple rule-based chatbot 4.3.2 Inference using forward chaining 4.3.3 Creating a simple rule-based chatbot using Python 4.3.3.1 Demonstrating forward chaining using a rule engine (e.g., PyKnow) 4.4 Case Study on Real-World AI Tools 4.4.1 Case Study 1: ChatGPT and Google Gemini – Conversational AI overview and its working principles 4.4.2 Case Study 2: Teachable Machine by Google – Building custom image/sound classification model without code 4.4.3 Case Study 3: IBM Watson – Understanding applications in healthcare, customer service, and business analytics [Students are expected to work on a Mini Project during lab hours designing AI based application Considering follows guide-liens: (i) Define a problem (e.g., chatbot, recommender, image classifier) (ii) Design and implement using Python or any AI tool (iii) Submit report with screenshots and working demo]
Reference Books	1) Artificial Intelligence: A Modern Approach, Stuart Russell, Peter Norvig, Pearson, ISBN: 978-0136042594 2) Python Programming: An Introduction to Computer Science, John Zelle, Franklin, Beedle & Associates Inc., ISBN: 978-1590282410 3) Data Structures and Algorithms in Python, Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser, Wiley, ISBN: 978-1118290279 4) Introduction to Artificial Intelligence and Expert Systems, Dan W. Patterson, Prentice Hall, ISBN: 978-0132550707 5) Python Machine Learning, Sebastian Raschka, Vahid Mirjalili, Packt Publishing, ISBN: 978-1789955750 6) Fundamentals of Artificial Intelligence, N. P. Padhy, Prentice Hall India, ISBN: 978-8120328805 7) Programming in Python 3, Mark Summerfield, Addison-Wesley, ISBN: 978-0321680563 8) Artificial Intelligence: Principles and Techniques, Nils J. Nilsson, Morgan Kaufmann, ISBN: 978-1558604674 9) Data Structures Using Python, Rance D. Necaise, Wiley, ISBN: 978-1118957321 10) Artificial Intelligence: A New Synthesis, Nils J. Nilsson, Morgan Kaufmann, ISBN: 978-1558604674 11) Python for Data Analysis, Wes McKinney, O'Reilly Media, ISBN: 978-1491957660 12) Machine Learning Using Python, Manaranjan Pradhan, BPB Publications, ISBN: 978-9388519004
Teaching Methodology	Class Work, Discussion, Lab work, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.